

SATD Detection: Qwen3 vs E5 Embeddings

What we did and what we found

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1 What we did

We took the SATD dataset (31 source files, code comments + issues + commits + pull requests), cleaned it up, and balanced the classes so no single label dominated. Then we ran everything through two embedding models, **Qwen3-Embedding** and **E5-Large**, and trained simple classifiers (**XGBoost** and **Logistic Regression**) on top of each, separately for:

- **Identification:** is it SATD or not?
- **Categorization:** if it's SATD, what type (C/D, DOC, TES, REQ)?

Then we compared all of that against the paper's BiLSTM/BERT numbers (and the replicated commit-message run) to see how well plain embeddings + simple classifiers stack up.

2 Results

Identification (SATD vs Not-SATD)

Artifact	Baseline	Qwen3+XGB	E5+XGB	Best embedding
Code Comments	0.939	0.915	0.891	below baseline
Issues	0.878	0.798	0.778	below baseline
Pull Requests	0.862	0.792	0.717	below baseline
Commit Messages	0.910	0.686	0.751	below baseline

Table 1: Macro F1. Embeddings consistently fall short here.

Categorization (C/D, DOC, TES, REQ)

Artifact	Baseline	Qwen3+XGB	E5+XGB	Best embedding
Code Comments	0.882	0.900	0.904	beats baseline
Issues	0.899	0.941	0.939	beats baseline
Pull Requests	0.876	0.927	0.936	beats baseline
Commit Messages	0.980	0.959	0.938	below baseline

Table 2: Macro F1. Embeddings win here for 3 of 4 artifact types.

3 Takeaways

- **XGBoost always beat Logistic Regression**, no exceptions, across all 32 models.

- **For identification**, the original BiLSTM still wins everywhere. Makes sense, since it's trained specifically for that task with augmented data.
- **For categorization**, our embeddings actually beat BERT for 3 out of 4 artifact types (code comments, issues, pull requests). Only commit messages lost, and that's against the very strong 0.98 replication.
- **Qwen3 vs E5**: pretty close overall. Qwen3 is slightly better for identification, E5 slightly better for categorization. Margins are small either way (2-3 points).
- **Overall best single combo**: Qwen3 + XGBoost (avg 0.865 across both tasks).

Bottom line: plain embeddings + XGBoost can't beat a purpose-trained BiLSTM for SATD identification, but they match or beat fine-tuned BERT for categorizing SATD types, and they're way cheaper to run since there's no fine-tuning involved.